Band-1 Hadamard Order 6212 — Miyamoto (Kronecker H_4×H_388 + C_1554) + Paley-I SPK Path; 6212 IN SPK via Paley-I; Gap 2052 Still OPEN

H_388/H_1552/C_1554/H_6212 Miyamoto PASS; H_6212 Paley-I PASS (q=6211); 6212 IN SPK via Paley-I (NOT Miyamoto-only); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_6196_Miyamoto`; `2026-09-15_Band1_Hadamard_5956_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_6212_miyamoto/`.
Cutoff. 2026-09-15T22:58:32Z (~6:58PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 388, 1552, and 6212 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Kronecker seed $H_{1552} = \text{Kronecker}(H_4, H_{388})$ where $H_{388} = \text{Paley-II}(q = 193)$ and symmetric Paley conference C_{1554} over prime $q = 1553 \equiv 1 \pmod 4$, **plus** the classical Paley-I path $H_{6212} = \text{Paley-I}(q = 6211)$ (prime $\equiv 3 \pmod 4$)). Seed triage for order 1552: Paley-II ($q = 775 = 5^2 \cdot 31$), Double, and Paley-I-direct ($q = 1551 = 3 \cdot 11 \cdot 47$) all **FAIL**. SPK audit for 6212: Paley-I **PASS**; Paley-II ($q = 3105 = 3^3 \cdot 5 \cdot 23$) **FAIL**; Kronecker **FAIL**. Therefore order **6212 is IN SPK via Paley-I** (Miyamoto is an alternate construction; **NOT** Miyamoto-only). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. `clay_closed = false`.

Abstract

Finite constructions of orders **388**, **1552**, and **6212**.

- H_{388} = Paley-II ($q = 193$)
- H_{1552} = Kronecker(H_4 Sylvester, H_{388})
- C_{1554} = Paley conference over prime 1553
- H_{6212} = Miyamoto(H_{1552}, C_{1554}) **and** Paley-I ($q = 6211$)

SPK: 6212 IN SPK via Paley-I (NOT Miyamoto-only). Seeds 388 and 1552 enter SPK. Prior Miyamoto-only bonuses (incl. 6196) remain excluded.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_6212.py`

Results. overall_pass=true; units 96/96; wall 403.5608s; all max|HH^T-nI|=0.

Cutoff: 2026-09-15T22:58:32Z (~6:58PM ET)

Arithmetic

- Target $n = 6212 = 4 \cdot 1553$; 1553 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1552} and C_{1554} (Paley conference over prime 1553).
- Seed: Kronecker(H_4, Paley-II H_{388} with $q = 193$); Paley-II/Double/Paley-I-direct for 1552 all FAIL.
- SPK: Paley-I $q = 6211$ **PASS** (prime $\equiv 3 \pmod 4$); Paley-II $q = 3105$ **FAIL**; Kronecker **FAIL → IN SPK via Paley-I**.

Constructions (verified)

Object	Method	max|HH^T-nI|	Wall (s)
H_4 Sylvester	Sylvester	0	0.0001
H_388 Paley-II	Paley-II q=193	0	0.0441
H_1552 Kronecker	Kronecker(H_4, H_388); triage FAIL Paley-II/Double/Paley-I-direct	0	1.7072
C_1554	Paley conf q=1553	0	2.7252
H_6212 Miyamoto	Miyamoto(H_1552, C_1554)	0	188.0334

H_6212 Paley-I	Paley-I q=6211 (SPK path)	0	206.2369

SPK closure note

Seeds include prior SPK ships through **6196** lineage (**1548** from that ship; **NOT** 6196 Miyamoto-only) + verified **388** / **1552** / **6212** (Paley-I); **exclude** Miyamoto-only bonuses including **6196/5956/5812/5476**. Known count 1282. Next constructive gap **2052** (OPEN).

Quantity	Value
6212 in SPK	**True** (Paley-I q=6211)
1552 in SPK	**True** (Kronecker 4×388)
388 in SPK	**True** (Paley-II)
6196 in SPK	**False** (prior Miyamoto-only)
Next constructive gap	**2052** (OPEN)

Next Miyamoto targets (after q=1553)

q	n=4q	need H	build notes
1597	6388	1596	[‘PaleyII(q=797)’]
1601	6404	1600	[‘Kronecker(4x400)’]
1609	6436	1608	[‘PaleyI(q=1607)’]
1613	6452	1612	[]
1621	6484	1620	[‘PaleyI(q=1619)’, ‘PaleyII(q=809)’]
1637	6548	1636	[]

Honesty

clay_closed = false · GATE-04 LOCKED · Hadamard conjecture OPEN · gap 2052 OPEN · no RH/P!=NP/YM/Frankl/NS/AC/Clay-closed claims.